

Lecture 9 - Part I

The firm - production

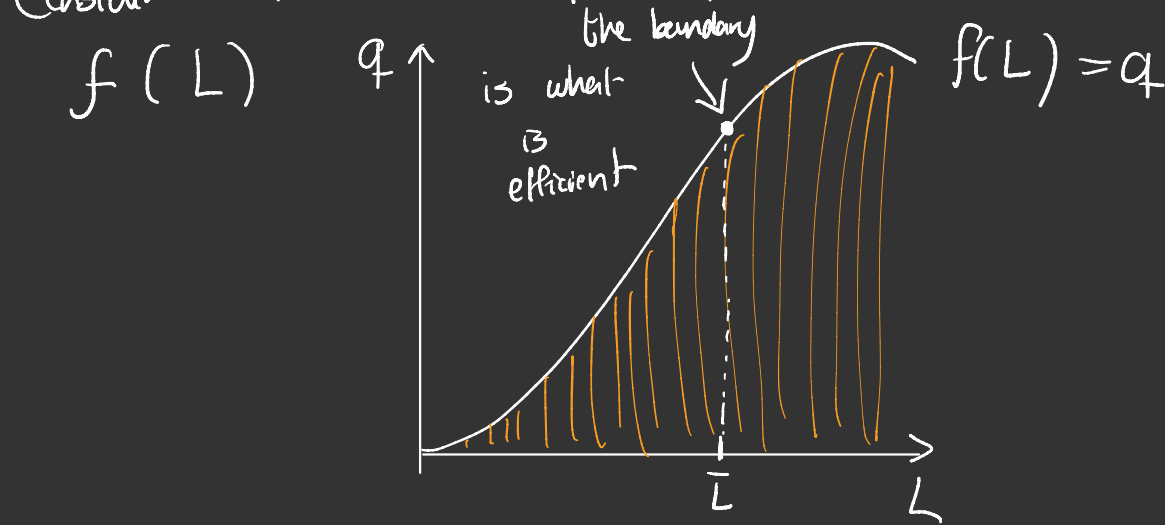
ECON 101
Fall 2020



Production function

Describes what is technically feasible and efficient to produce

Consider this one-input production function of labor

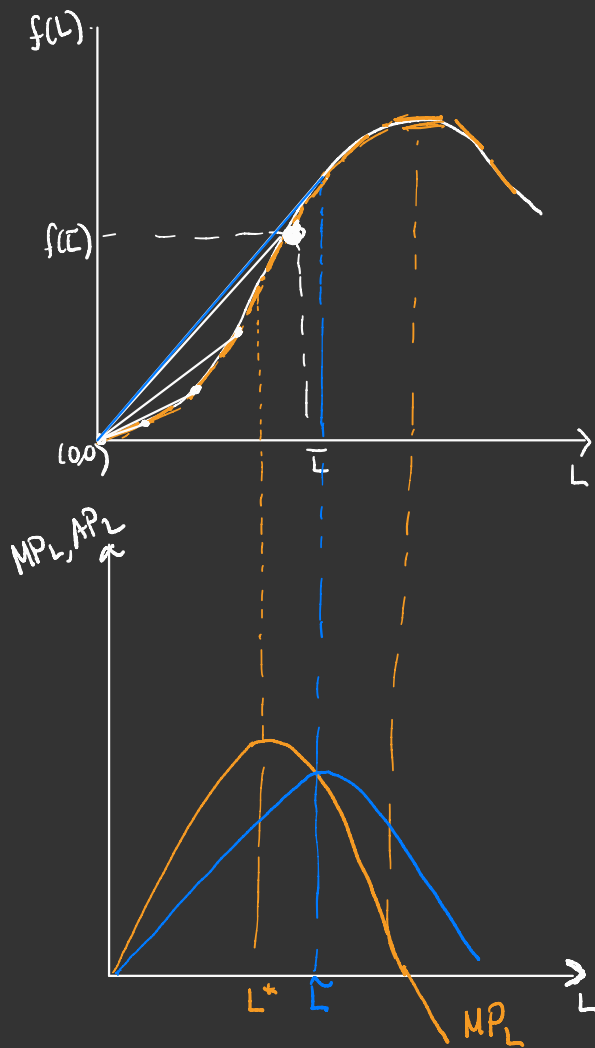


Average and Marginal Products

$$q = F(K, L),$$

$$\left. \begin{array}{l} \text{Average Product for Labor} = AP_L = \frac{q}{L} \\ \text{Marginal Product for Labor} = MP_L = \frac{\Delta q}{\Delta L} \end{array} \right\} \text{while } K = \bar{K} \text{ fixed}$$

$$\left. \begin{array}{l} AP_K = \frac{q}{K} \\ MP_K = \frac{\Delta q}{\Delta K} \end{array} \right\} \text{while } L = \bar{L} \text{ fixed.}$$



Deriving MP_L and AP_L graphically

$$MP_L = \frac{\Delta q}{\Delta L} = \frac{\Delta f(L)}{\Delta L} \approx \text{slope of } f(L)$$

slope tangent line at each L

$$AP_L = \frac{q}{L} = \frac{f(L)}{L}$$

Consider a secant line $L=0, q=0$
 $\bar{L}, f(\bar{L})$

↓

$$\text{slope of this secant line} = \frac{f(\bar{L}) - 0}{\bar{L} - 0} = AP_L(\bar{L})$$

MP_L reaches a maximum before AP_L does

MP_L crosses AP_L at maximum of AP_L

L^+ is the labor that gives the max MP_L

$\tilde{L}$ is the labor that gives the max AP_L

$MP_L(\tilde{L}) = AP_L(\tilde{L})$ and only at that point

Two-variable input production

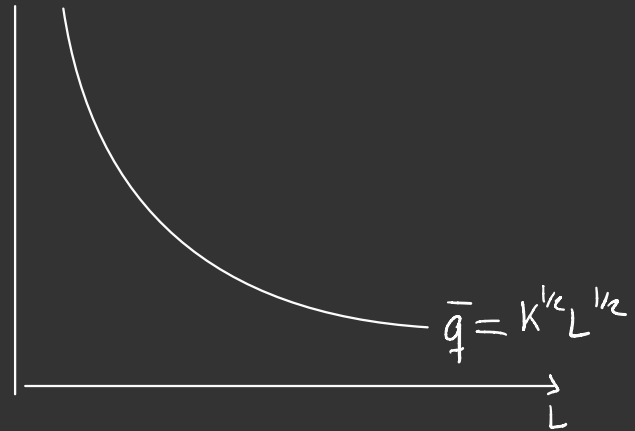
$$q = F(K, L)$$

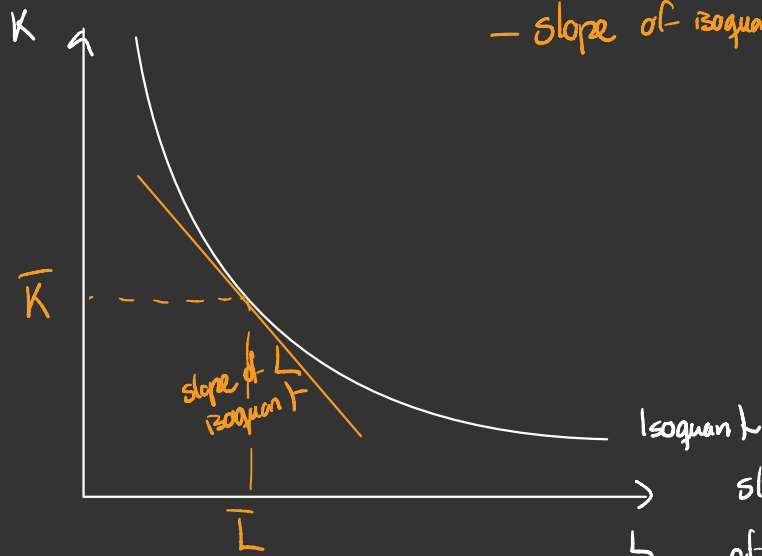
an iso-quant is the set of input combinations (K, L) that yield the same level of output $\bar{q}$:

implicitly given by the eq. $\bar{q} = F(K, L)$

For example: $\bar{q} = A K^{1/2} L^{1/2} \Rightarrow \left(\frac{\bar{q}}{A L^{1/2}} \right)^2 = K$

$\Rightarrow \frac{\bar{q}^2}{A^2 L} = K$ eq. isoquant corresponding to prod. level $\bar{q}$





— slope of isoquant $= - \frac{\Delta K}{\Delta L} = MRTS$ marginal rate of technical substitution

$q = F(K, L) \Rightarrow$ Change input combination by $\Delta K, \Delta L$

$$\Rightarrow 0 = MP_K \Delta K + MP_L \Delta L$$

$$\text{slope} \approx \frac{\Delta K}{\Delta L} = - \frac{MP_L}{MP_K} = -MRTS$$

of isoquant

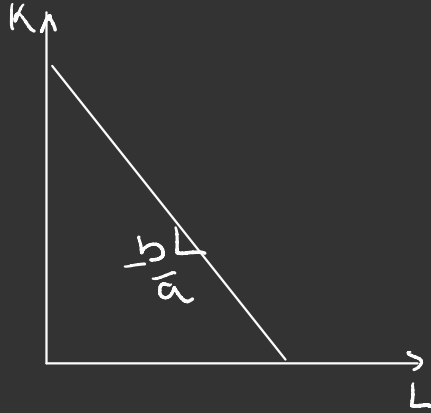
$$\begin{aligned} MRTS &= |\text{slope of isoquant}| \\ &= \frac{MP_L}{MP_K} \end{aligned}$$

Three examples

$$q = aK + bL$$

Perfect Substitutes

$$MRTS = \frac{b}{a}$$



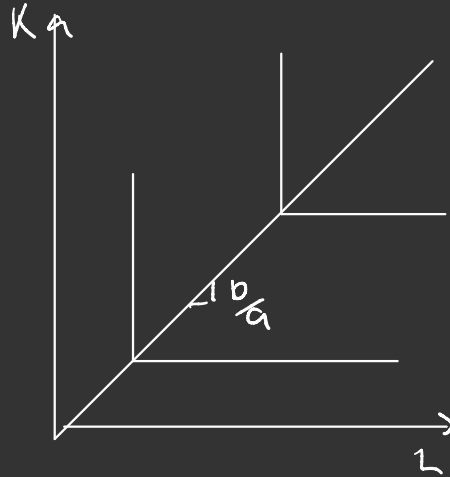
Perfect Substitutes

$$q = \min \{aK, bL\}$$

$$\text{Kink} \Rightarrow aK = bL$$

$$\Rightarrow K = \frac{b}{a}L$$

$MRTS = \text{undefined}$

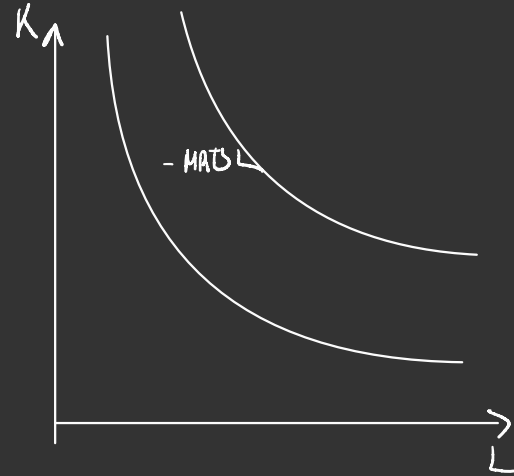


Fixed Proportions
Leontief production
function

$$q = A K^{\alpha} L^{\beta}$$

Cobb-Douglas production
function

$$MRTS = \frac{\alpha}{\beta} \cdot \frac{K}{L} \quad (L \text{ vs } K)$$



Returns to scale

For one-input production function $q = f(L)$

Increase input by the
constant λ

$$f(\lambda L) = \lambda^k f(L) = \lambda^k q$$

↑
if I'm able to
factor like this

$f(L)$ exhibits some
kind of returns to scale

$$\left\{ \begin{array}{ll} k=1 & \text{CRS} \\ k>1 & \text{IRS} \\ k<1 & \text{DRS} \end{array} \right.$$

$$f(L) = L \Rightarrow f(\lambda L) = \lambda L = \lambda f(L) : \text{CRS}$$

$$f(L) = \sqrt{L} \Rightarrow f(\lambda L) = \sqrt{\lambda} \sqrt{L} = \lambda^{1/2} f(L) : \text{DRS}$$

$$f(L) = L^2 \Rightarrow f(\lambda L) = \lambda^2 L^2 = \lambda^2 f(L) : \text{IRS}$$

$$f(L) = \frac{1}{10} (10L^2 - L^3)$$

There are prod.

functions that cannot

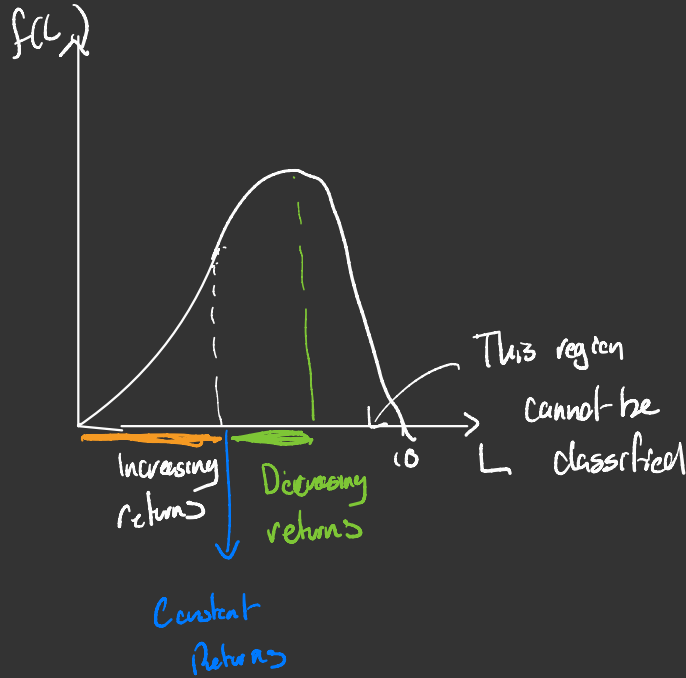
be classified so easily...

$$f(\lambda L) = \frac{1}{10} (10(\lambda L)^2 - (\lambda L)^3) = \frac{1}{10} (\lambda^2 10L^2 - \lambda^3 L^3)$$

can't
be
factored

$$\text{as } f(\lambda L) \neq \lambda^k f(L)$$

$$f(L) = \frac{1}{10} (10L^2 - L^3)$$



Two-Input Production functions:

$$q = F(K, L)$$

If I increase factors to $\lambda K, \lambda L$, i.e., by λ :

$$F(\lambda K, \lambda L) = \lambda^K F(K, L) = \lambda^K q$$

If I can do this factorization then we can classify the prod. function

$$- K = 1 \quad \text{CRS}$$

$$- K > 1 \quad \text{IRS}$$

$$- K < 1 \quad \text{DRS.}$$

If I can't perform the factorization, then classification is not as easy.

Example: $q = A K^\alpha L^\beta = F(K, L)$

Increases $(K, L) \rightarrow (\lambda K, \lambda L) : A(\lambda K)^\alpha (\lambda L)^\beta = \lambda^{\alpha+\beta} A K^\alpha L^\beta = \lambda^{\alpha+\beta} F(K, L)$

$\Rightarrow F(\lambda K, \lambda L) = \lambda^{\alpha+\beta} F(K, L) = \lambda^{\alpha+\beta} q$

Therefore :

- If $\alpha + \beta = 1$ CRS
- If $\alpha + \beta > 1$ IRS
- If $\alpha + \beta < 1$ DRS